CoCo Zero

The standard, version 2.1

A CoCo Zero is a Tandy Color Computer emulator on a board the size of a Raspberry Pi Zero. Switch it on and BASIC is there, with no operating system in between. It is at least a 128K CoCo 3. It plugs into an ordinary TV with a mini HDMI cable, takes a USB keyboard and gamepad, and keeps its disks and programs on a microSD card. Anyone can build one, on any board of that size, and every one has the same outline and mounting holes.

A CoCo Zero is anything that meets every requirement on this page.

Form factor

A CoCo Zero is built on a board of the Raspberry Pi Zero size.

All dimensions in millimeters. The origin is one corner of the board, with X along the 65 mm edge and Y along the 30 mm edge.

Feature	Value
Board outline	65 x 30
Corner radius	3.0
Mounting holes	4, for M2.5, drilled 2.75 +/- 0.05
Hole centers	(3.5, 3.5), (61.5, 3.5), (3.5, 26.5), (61.5, 26.5)

Machines and memory

- A CoCo Zero emulates a CoCo 3 with at least 128K of memory.
- A CoCo 1 or CoCo 2, where one is offered, has at least 64K of memory.

Zero operating system

- The firmware runs on the bare hardware, with no operating system.
- Libraries are not operating systems. A library the firmware is built on is allowed.
- Alongside the emulated CoCo there is one configuration screen. It may:
  - set the emulator's options;
  - mount disk images;
  - show system information;
  - load programs and cartridges directly.
- The firmware runs nothing else.

Features

Connectors

- Two USB-C ports:
  - one for power and programming;
  - one a USB host for peripherals.
- Video output is a mini HDMI connector.
- Storage is a microSD card slot.

USB devices

- A standard USB-C keyboard works when plugged into the host port.
- A USB joystick or gamepad works as the CoCo's joysticks.
- A keyboard and a joystick work together through a simple USB hub (one with a single hub chip).
- A class-compliant USB audio device can play the sound in place of the video cable.

Video and sound

- 640 x 480 at 60 Hz, as a DVI signal.
- Sound is embedded in the DVI signal.

Performance

- The emulated CoCo runs in real time at double speed (about 1.79 MHz).
- No dropped frames.
- No gaps in the sound.
- Power-on goes straight to the BASIC prompt.

Cooling

- Passively cooled. No fan.

Firmware

- The firmware is open source.

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